

GABLS1: SLD $\gamma=4$

GABLS1 stable surface-layer diffusivity, $\gamma=4$ | 23 September 2026. A matched nine-hour, 32-cubed, 12.5 m WENO9 case changes only the interior SLD stable-similarity strength from $\gamma=1$ to $\gamma=4$. Neutral $\gamma=0$ and filtered no-closure are admitted controls. The fixed 1 m LES median is an intercomparison reference, not observations.

During 8-9 h, first-layer vector shear is 0.12911 per second at $\gamma=4$, versus 0.05302 at $\gamma=0$, 0.07387 at $\gamma=1$ and 0.10038 for the reference sampled at the same heights. $\gamma=4$ overshoots the reference by 0.02873 per second; $\gamma=1$ undershoots by 0.02651. The two strengths bracket the median, with nearly equal absolute shear errors. The 7-8 h $\gamma=4$ shear is 0.11921 per second.

Peak resolved w -squared during 8-9 h is 0.08575 square meters per second squared, up from 0.06276 for $\gamma=0$ and 0.06676 for $\gamma=1$. At 18.75 m, w -cubed is +0.005932 cubic meters per second cubed, versus -0.000295 and +0.000102. These moments are closer to the reference profile, while the overshoot mean shear limits the overall improvement. $\gamma=4$ w -squared and w -cubed rose between 7-8 and 8-9 h, so statistical equilibration remains uncertain.

At the first interior face, mean $\gamma=4$ local similarity functions over 8-9 h are $\phi_m=2.7169$ and $\phi_h=3.7900$. Every saved column is on the stable branch, with no upward-flux fallback. The evolved viscosity is 0.24517 square meters per second, versus 1.2892 at $\gamma=0$ and 0.82387 at $\gamma=1$. Heat diffusivity is 0.088672 versus 1.2396 and 0.60453. The wall law and filters are unchanged.

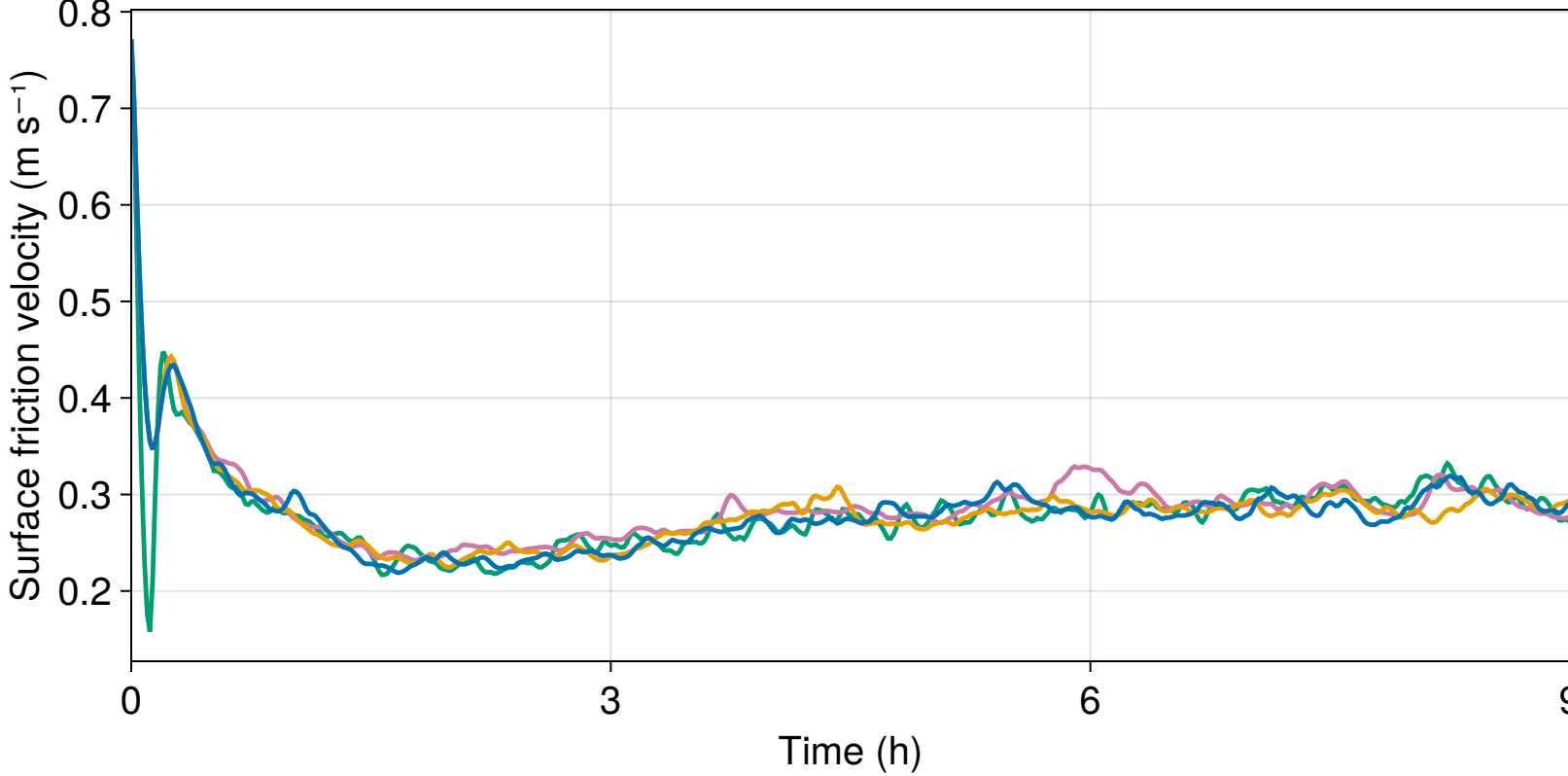
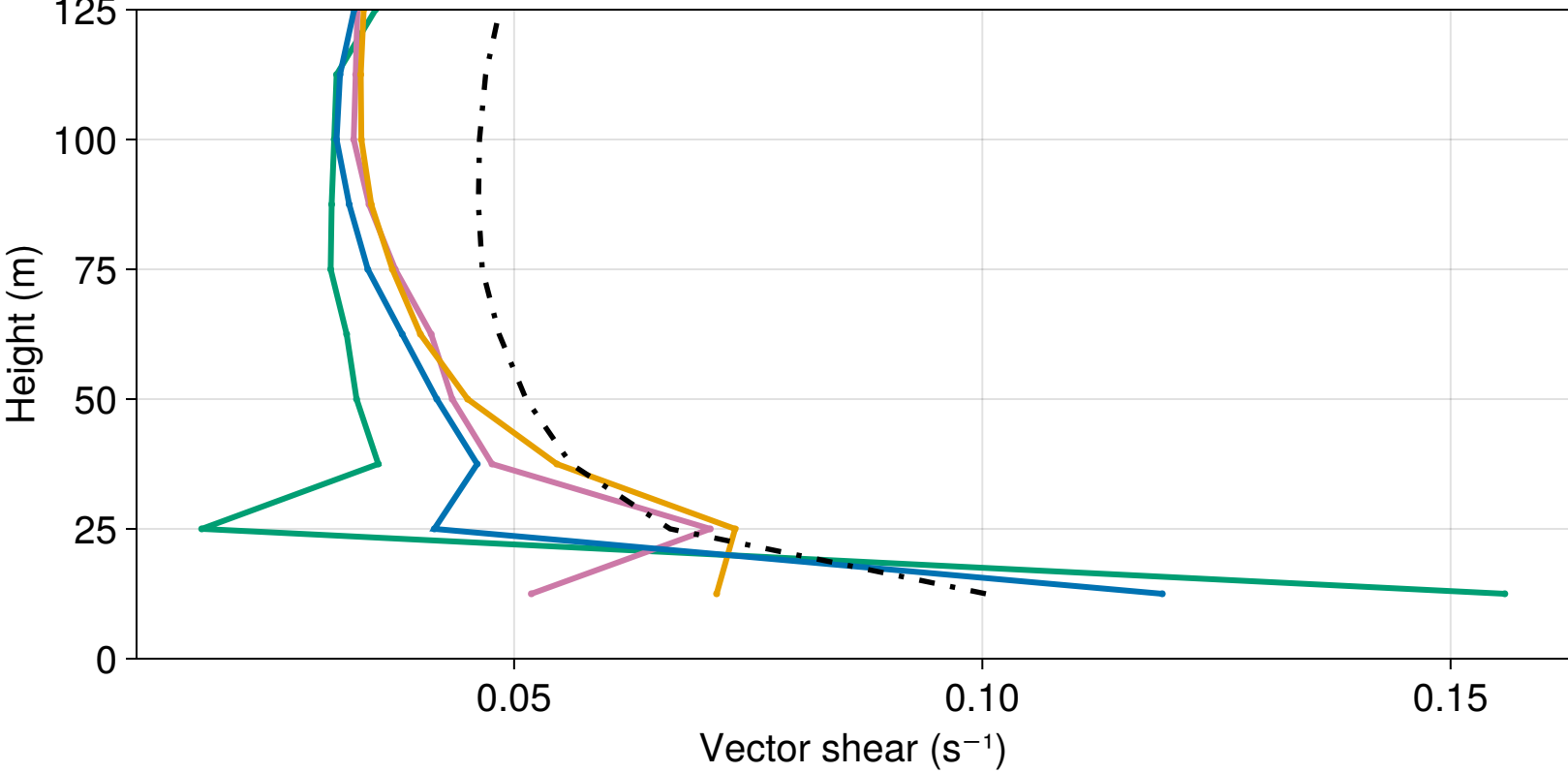
Reduced closure changes the transport partition: first-face resolved $u-w$ is -0.045186 and SGS $u-w$ is -0.027868 square meters per second squared at $\gamma=4$, compared with -0.014608 and -0.052218 at $\gamma=1$. Resolved w -theta becomes -0.010566 and SGS w -theta -0.001608 kelvin meters per second. The measured WENO u -flux correction is -0.001743, tracked separately from covariance.

Validation: CPU smoke 7583 passed 48/48; H100 gate 7584 passed CUDA parity 34/34 and the 1800 s case 48/48. Science 7585 ended with durable exit zero and CASE_DONE at 32400 s. Independent Julia audits verified the immutable source manifest, paired initial-theta digest, 46 native-height profiles, 135 series, finite values, exact schedules, filter evolution, density-consistent wall flux, local inverse Obukhov length and similarity identities at 55 times, and reconstructed transport accounting. The result is one seed at one coarse resolution, not a universal parameter calibration.

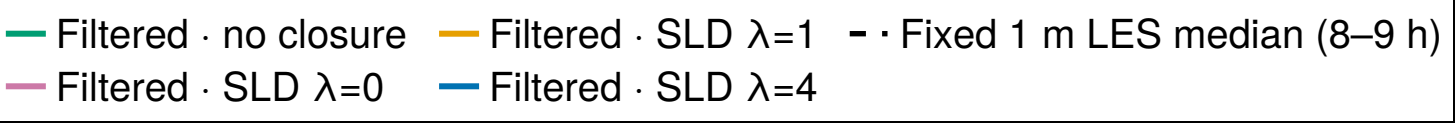
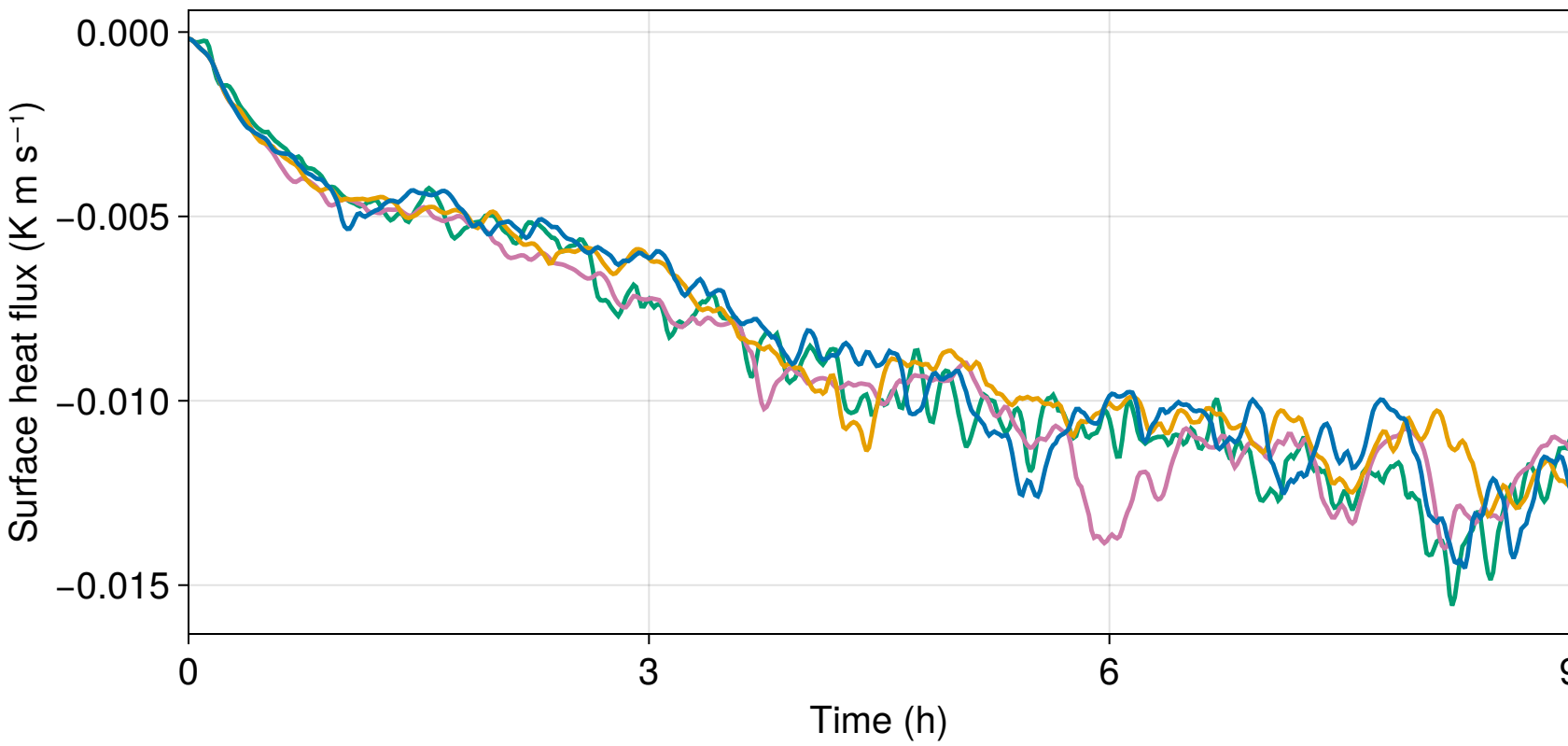
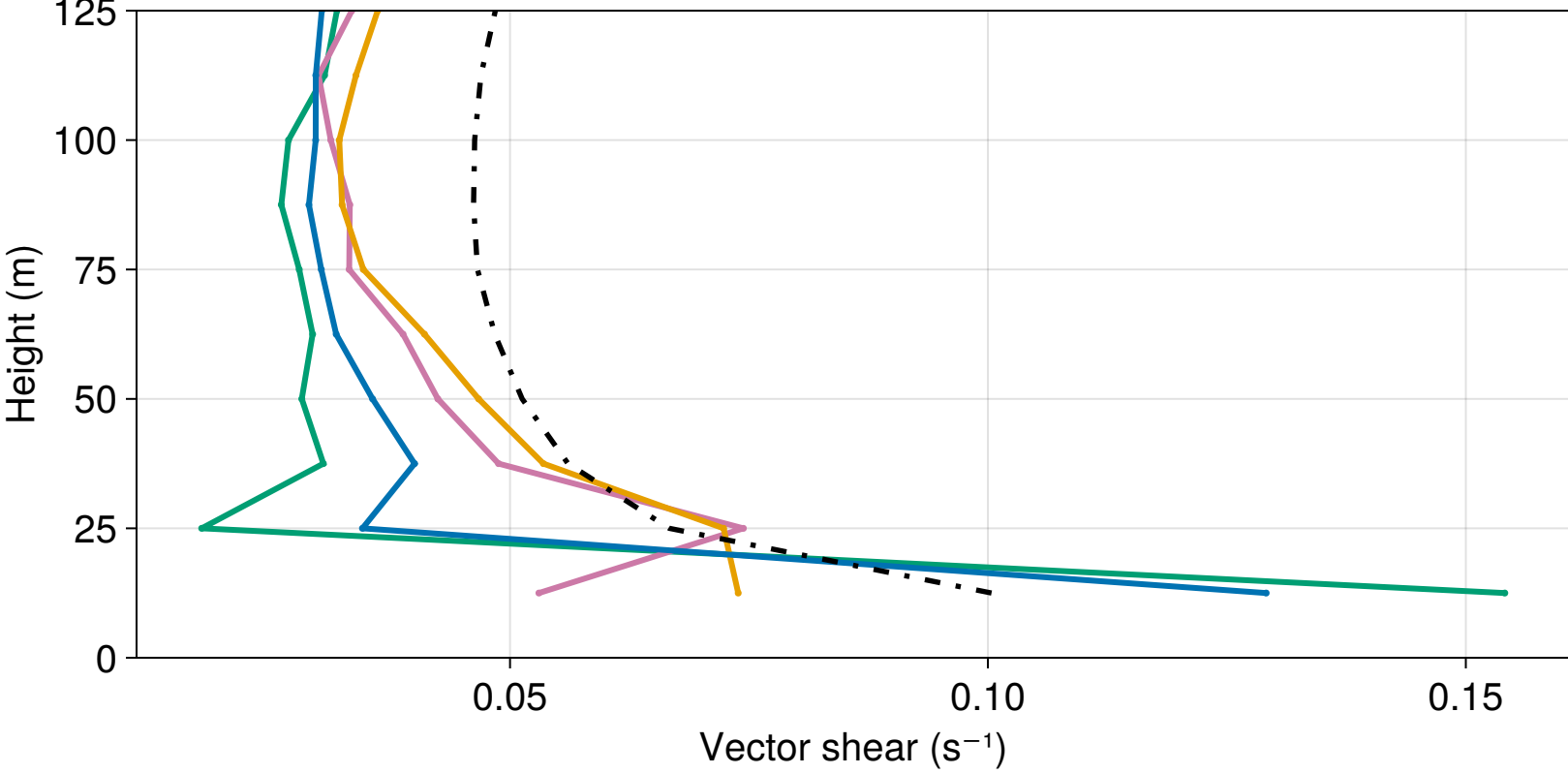
Mean-wind shear and wall flux

Shear between adjacent 12.5 m cell centers; fixed 1 m median sampled at the same heights

7–8 h

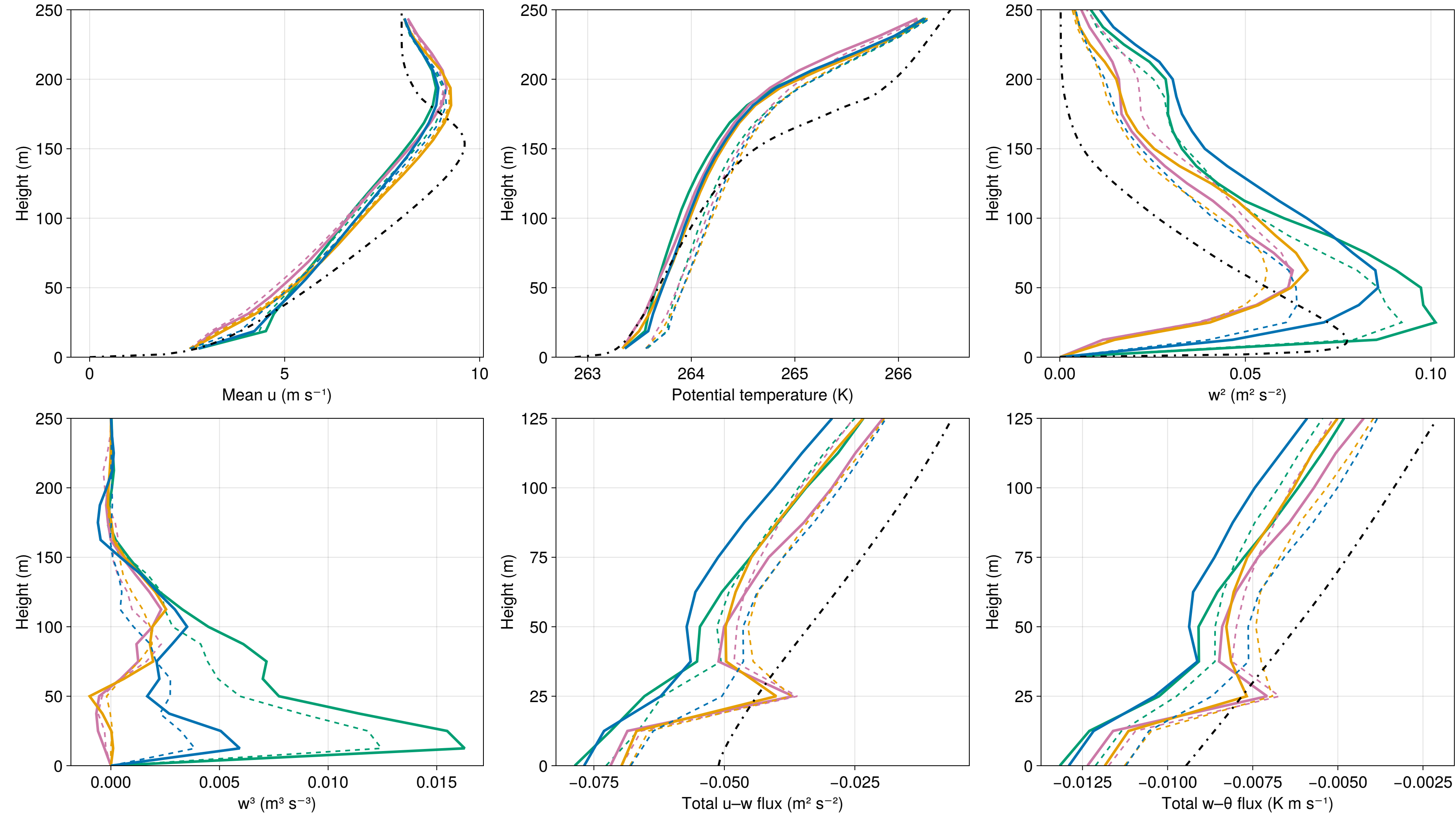


8–9 h



GABLS1: stable Monin–Obukhov correction of SLD

32³ · 12.5 m · WENO9 · filtered wall · scheme-native one-face SLD · $\lambda=0, 1, 4$ · dashed 7–8 h, solid 8–9 h

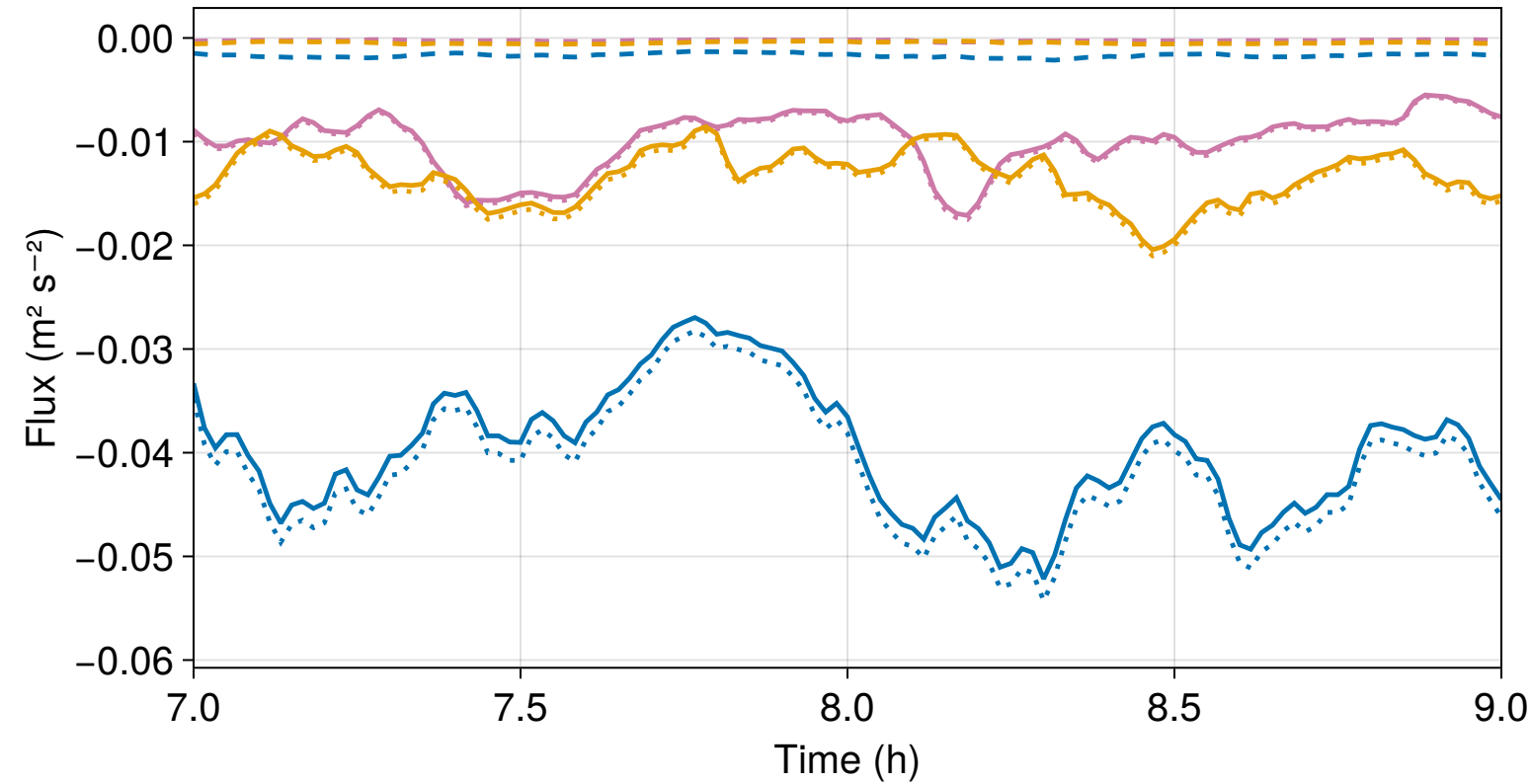


Filtered · no closure Filtered · SLD $\lambda=0$ Filtered · SLD $\lambda=1$ Filtered · SLD $\lambda=4$ - · Fixed 1 m LES median

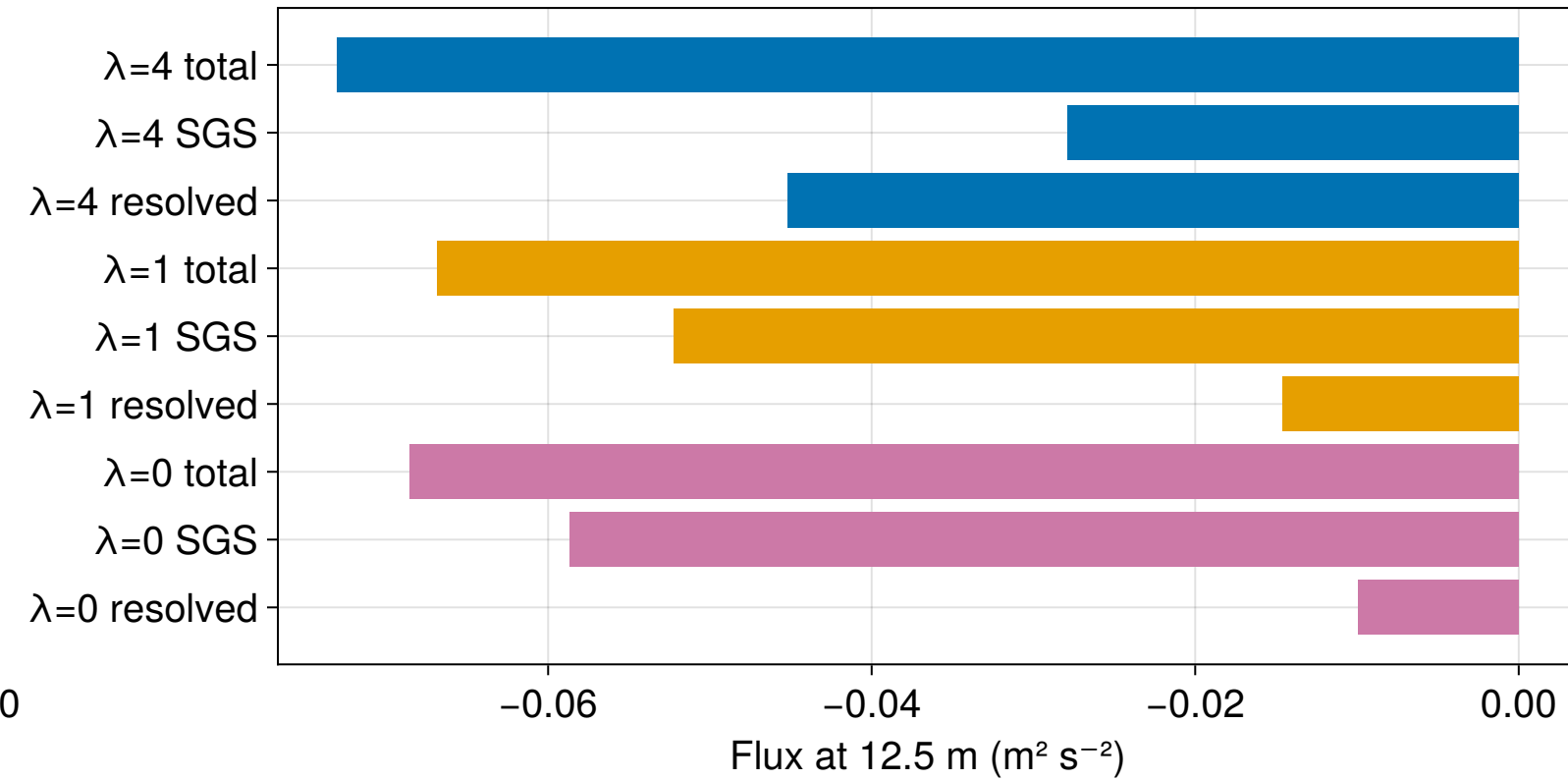
Scheme-native SLD: first-face transport partition

Reconstructed resolved = covariance + numerical correction; SGS is the SLD constitutive flux

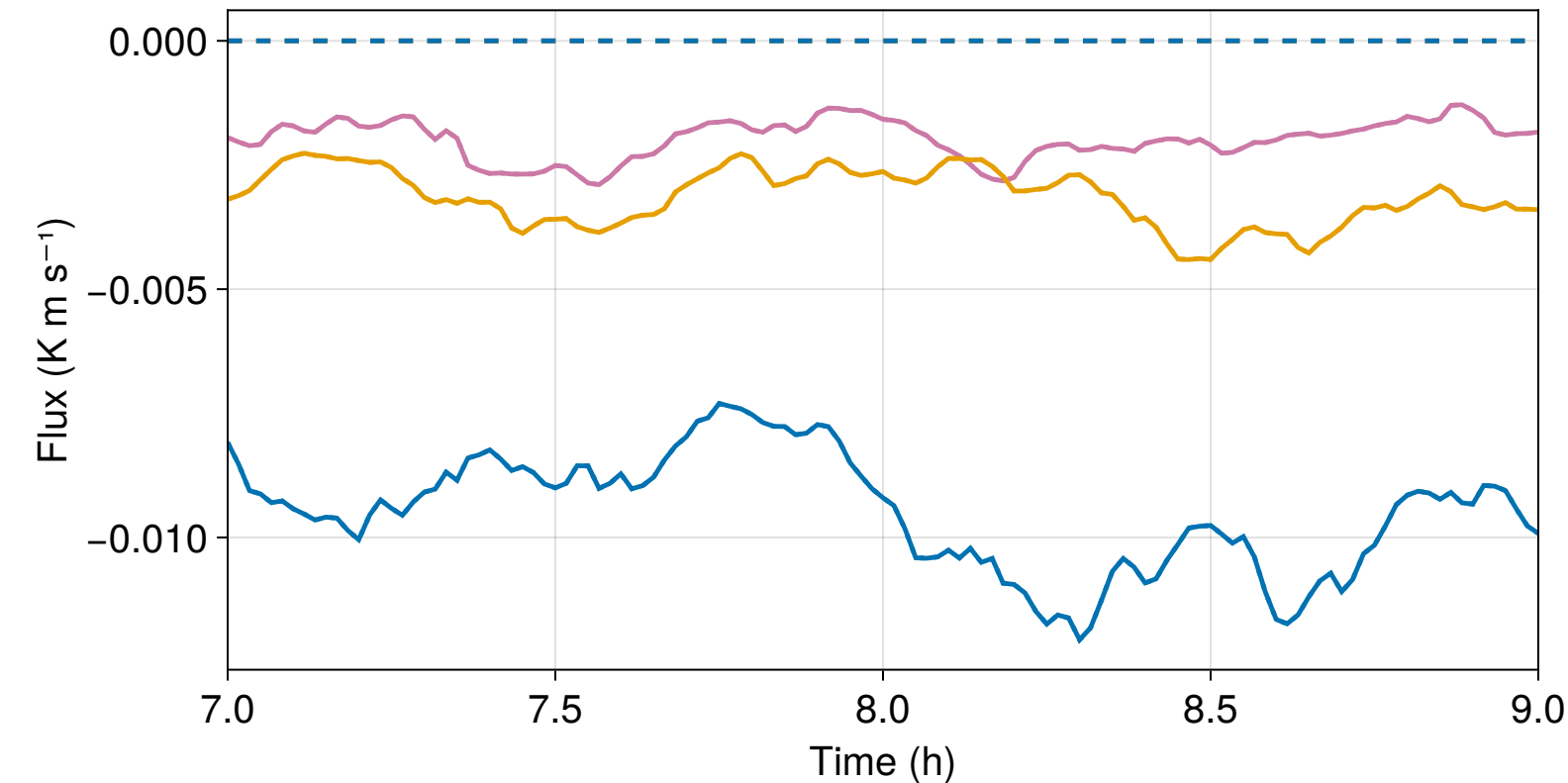
Momentum



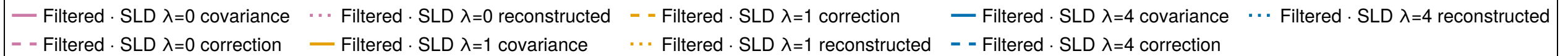
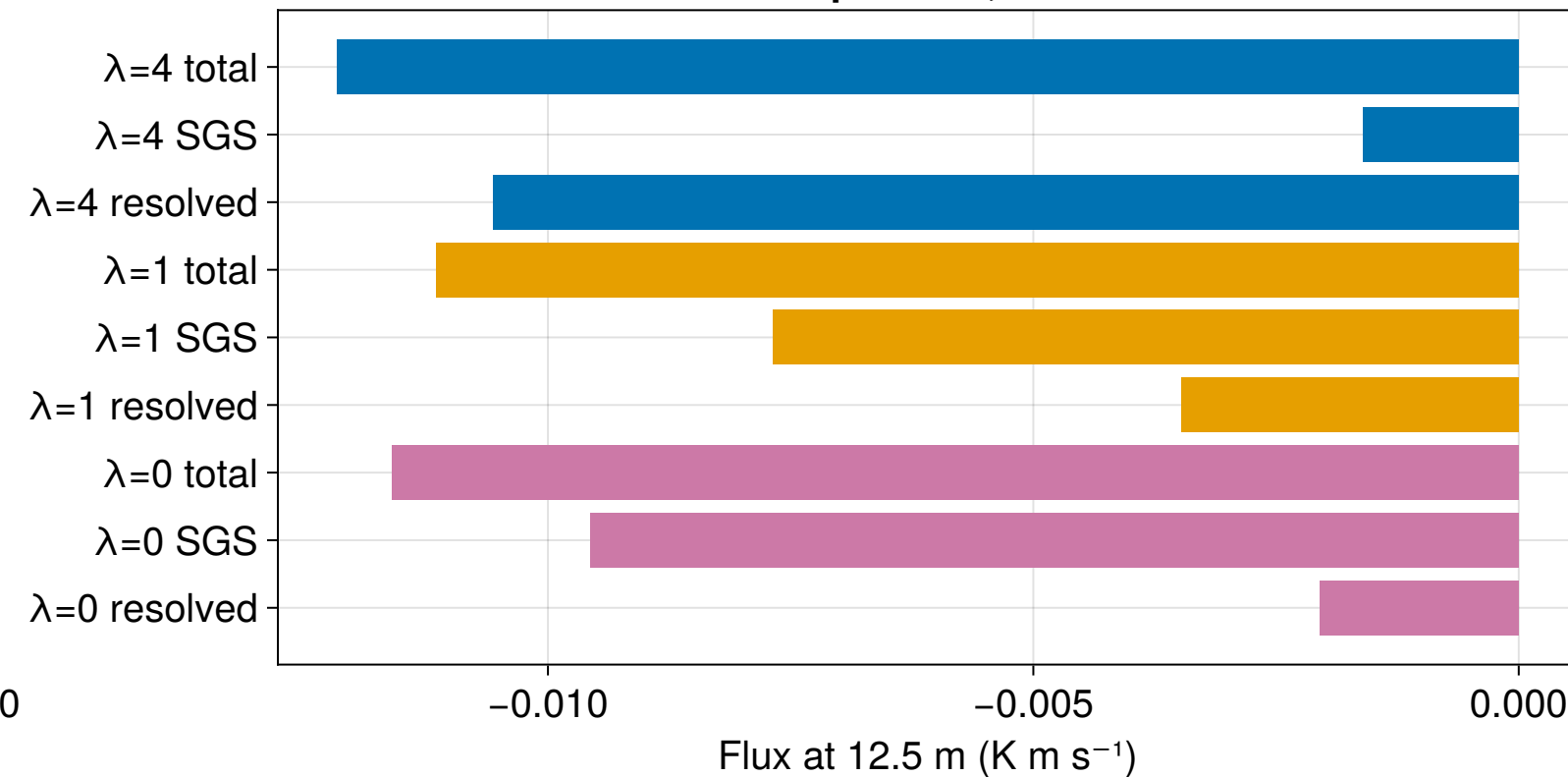
u-w partition, 8–9 h



Heat



w-θ partition, 8–9 h



First-face SLD coefficients and local stability

Horizontal means of local quantities at $z = 12.5$ m; φ is never evaluated from a mean L

